

Sparse Congruence-Preserving Perturbations of Dyadic Series

What gap statistics can tell us about Erdős Problem #251

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ABSTRACT

For each convergent dyadic series with nonnegative integer coefficients, we construct non-negative integer perturbations whose sums fill an interval, with one sparse permitted set for all targets. The corrections obey any prescribed bound tending to infinity and eventually preserve coefficient and partial-sum residues modulo every fixed integer. Pairs with a fixed ordinary sum provide the weighted choices for interval covering. For prime gaps we also preserve growing-block statistics and cumulative size. The cumulative positions need not be prime, and the irrationality question remains open.

1 Introduction

Let $p_0 = 2, p_1 = 3, \dots$ be the primes and put $g_n = p_{n+1} - p_n$. Erdős asked whether

$$\Pi = \sum_{n \geq 0} \frac{p_n}{2^{n+1}}$$

is irrational [1, p. 94][2, p. 62] [3, p. 103]. Summation by parts gives $\Pi = 2 + \sum_{n \geq 0} g_n 2^{-(n+1)}$, with convergence justified in Section 4. We study which properties of the gaps survive when the latter sum is changed to a prescribed value.

We prove a perturbation theorem for arbitrary convergent dyadic series with non-negative integer coefficients. A single sparse set of permitted indices allows every target in an interval, even when the corrections must obey an arbitrarily slowly growing bound. We also prescribe, before choosing the target, a cutoff for each modulus beyond which both the corrections and their partial sums are divisible by that modulus. For prime gaps the same construction retains the distributions of blocks of length $o(\log \log X)$ sampled in $[X, 2X)$ and gives cumulative positions asymptotic to $n \log n$. The irrationality of the actual prime series remains a distinct question: these comparison sequences need not have prime cumulative positions.

We use a standard interval-covering argument for series with finite choices. Fridy's generalised-base lemma [5, p. 194] treats bounded digits and decreasing weights. Cr-marić and Kovač [17, Lemma 4] give the finite-choice form used here, and Kovač and

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Tao [6, Lemma 5.1] use analogous intervals of reciprocal choices. We must arrange these choices on a sparse set while preserving cumulative congruences. To do so, we vary two adjacent corrections with a fixed ordinary total. For example, the pairs $(0,6), (2,4), (4,2), (6,0)$ at indices $n, n+1$ contribute $6, 8, 10, 12$ divided by 2^{n+2} , and each pair has total 6. A correction at $n-1$ first fixes the cumulative residue, after which the choice of pair leaves all later residue corrections unchanged. Sparsity requires the triples to move apart. We choose their spacing and the range of each pair together, so that later choices still cover the gaps between the current ones.

Throughout, $\mathbb{N} = \{0, 1, \dots\}$, intervals of indices contain integers, and $\log$ is natural unless a base is displayed. A set $S \subseteq \mathbb{N}$ has *upper Banach density zero* if

$$\lim_{H \rightarrow \infty} \sup_{u \in \mathbb{N}} \frac{|S \cap [u, u+H)|}{H} = 0.$$

For integers $X, m \geq 1$, let $\mu_{a,X,m}$ denote the distribution of $(a_n, \dots, a_{n+m-1})$ when n is uniform on $[X, 2X)$. Blocks are unnormalised and counted with multiplicity. We use $d_{\text{TV}}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|$.

Proposition 1.1 (sparse changes preserving congruences). *Let $a : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $A = \sum_{n \geq 0} a_n 2^{-(n+1)} < \infty$, let $K \in \mathbb{N}$, and let $f : \mathbb{N} \rightarrow \mathbb{R}$ tend to $+\infty$. There exist a set $S \subseteq [K, \infty)$ of upper Banach density zero and a nondegenerate interval $I \subset (A, \infty)$ such that, for every $r \in I$, there is an integer correction $e : \mathbb{N} \rightarrow \mathbb{N}$ satisfying*

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$$\text{supp } e \subseteq S, \quad e_n \leq f(n) \text{ eventually}, \quad \sum_{n \geq 0} (a_n + e_n) 2^{-(n+1)} = r.$$

For each integer $q \geq 1$ there is a cutoff N_q , independent of r , such that $q \mid e_n$ and $q \mid \sum_{i < n} e_i$ for all $n \geq N_q$. For every $\varepsilon > 0$, the choice $f(n) = (\log(n+3))^\varepsilon$ can be made with

$$|S \cap [X, 2X)| = O_\varepsilon\left(\frac{X}{\log \log X}\right), \quad d_{\text{TV}}(\mu_{a,X,m}, \mu_{a+e,X,m}) \leq \frac{m|S \cap [X, 2X+m)|}{X}.$$

In particular the distance tends to zero for every integer-valued $m = m(X) \geq 1$ with $m(X) = o(\log \log X)$, uniformly over target values.

The support of e may depend on r and occupy only part of S . Polynomially growing nonnegative integer sequences satisfy the convergence hypothesis, whereas $a_n = 2^n$ does not. The allowance may grow as slowly as $\log \log(n+3)$. A bounded allowance would force every correction to vanish: a modulus exceeding the bound makes e eventually zero, and its constant cumulative sum must then be divisible by every positive integer.

Section 2 gives the construction, and Section 3 applies it to prime gaps. We then identify the actual tails and characterise their rationality by integral shifts. This leads to finite separation tests and to a sufficient condition involving two consecutive small differences. The [companion's literature discussion](#) contains the further comparisons, including finite subsums and other dyadic coefficient sequences.

2 Proof of Proposition 1.1

Proof. We first arrange the congruences so that choosing a weighted contribution will not change any later residue correction. We then choose the spacing and moduli to satisfy the size bound and fill an interval.

Let $n_j, j \geq 0$, be centres following an initial index $n_{-1} \geq K$, with separations $s_j = n_j - n_{j-1} \geq 4$. We use positive integers M_j satisfying $M_j \mid M_{j+1}$ and allow digits up to $D_j = 2^{s_j} - 1$. This range will compensate for the factor 2^{s_j} lost by moving the next choice farther out. If C_j is the total correction before $n_j - 1$, we clear its residue modulo M_j by setting, from $C_0 = 0$,

$$c_j = (-C_j) \bmod M_j \quad (0 \leq c_j < M_j), \quad C_{j+1} = C_j + c_j + M_j D_j.$$

For a digit $0 \leq d_j \leq D_j$, define

$$e_{n_j-1} = c_j, \quad e_{n_j} = M_j d_j, \quad e_{n_{j+1}} = M_j(D_j - d_j), \quad (1)$$

and set $e_n = 0$ elsewhere. Each triple has total $c_j + M_j D_j$, independently of d_j , so the recursion determines all C_j and c_j before we choose any digits. After the entry at $n_j - 1$ has cleared the residue, the two multiples of M_j preserve it. Thus the cumulative sum is divisible by M_j from index n_j through the end of the triple.

For a fixed $q \mid M_j$, the cumulative sum is divisible by q at n_j . At each later triple, divisibility of C_j and M_j implies $q \mid c_j$, and both entries of the pair are again multiples of q . Induction through the triples and the intervening zeros gives $q \mid e_n$ and $q \mid \sum_{i < n} e_i$ for every $n \geq n_j$. The entry c_j itself lies at $n_j - 1$, outside this range. We therefore need moduli eventually divisible by every fixed q , with spacings that leave enough weighted choices for interval covering.

To fit even a nonmonotone allowance, we replace it by the lower envelope

$$h(n) = \inf_{m \geq n} \min(f(m), m).$$

Since $f(n) \rightarrow \infty$, this envelope is finite, nondecreasing and tends to infinity. The additional bound $h(n) \leq n$ will ensure summability. We choose n_{-1} with $h(n_{-1}) \geq 32$ and set recursively

$$k_j = \max\{k \geq 2 : k!2^{k+2} \leq h(n_{j-1})\}, \quad M_j = k_j!, \quad s_j = k_j + 2, \quad n_j = n_{j-1} + s_j.$$

The set defining k_j is nonempty because $2!2^4 = 32$, and it is finite because factorials tend to infinity. As h is nondecreasing and $n_j \rightarrow \infty$, the integers k_j are nondecreasing and tend to infinity. Hence every fixed modulus eventually divides M_j , while $s_j \rightarrow \infty$. At a coordinate n of the j th triple, every entry of (1) is bounded by $M_j 2^{s_j} \leq h(n_{j-1}) \leq \min(f(n), n)$. This proves the required correction bound and convergence of all the weighted correction sums.

Let S be the union of these triples. Given R , we discard the finitely many centres preceding the point where all separations are at least R . Any interval of length H then meets at most $3(H/R + 2)$ permitted indices, in addition to a fixed finite set. We may

take the supremum over translates, let $H \rightarrow \infty$, and then let $R \rightarrow \infty$. Thus S has upper Banach density zero.

We now calculate the range of possible weighted sums. Increasing d_j by one transfers M_j from index $n_j + 1$ to index n_j , so it increases the dyadic contribution by $w_j = M_j 2^{-n_j-2}$. Write

$$\beta = \sum_{j \geq 0} (c_j 2^{-n_j} + D_j w_j), \quad F_j = \sum_{i \geq j} D_i w_i.$$

Here β is the contribution with all digits zero, and F_j is the sum of the maximal digit contributions from stage j onwards. The bounds above give $\beta > 0$, $0 < F_0 < \infty$ and $F_j \rightarrow 0$. To represent every value in that range, the later choices must cover the spacing w_j between successive current choices. Our definition of D_i was chosen for the identity

$$D_i w_i = M_i (2^{-n_{i-1}-2} - 2^{-n_i-2}).$$

For $i > j$, replacing M_i by M_j gives a lower bound whose sum telescopes. Since $n_i \rightarrow \infty$, we obtain

$$F_{j+1} \geq M_j 2^{-n_j-2} = w_j. \quad (2)$$

The intervals

$$[dw_j, dw_j + F_{j+1}], \quad d = 0, \dots, D_j,$$

therefore overlap or meet at their endpoints. Their union is $[0, F_j]$, because $F_j = D_j w_j + F_{j+1}$. Given $x \in [0, F_0]$, we choose d_0 so that $x - d_0 w_0 \in [0, F_1]$ and repeat this choice for each successive remainder. After stage j the remainder belongs to $[0, F_{j+1}]$, which shrinks to zero, and hence $x = \sum_j d_j w_j$. Adding the fixed contribution β proves the interval assertion with $I = (A + \beta, A + \beta + F_0)$. This is the finite-choice covering argument of [17, Lemma 4]. In particular, the permitted set, the interval and every congruence cutoff were chosen before the target x .

For the quantitative assertion, we choose separations of order $\log \log n$ so that the number of triples can also be bounded. Replace the preceding schedule by

$$s_j = \left\lfloor \frac{\varepsilon}{2} \log_2 \log(n_{j-1} + 3) \right\rfloor, \quad k_j = \max\{k \geq 2 : k! \leq (\log(n_{j-1} + 3))^{\varepsilon/4}\},$$

and again put $M_j = k_j!$, $n_j = n_{j-1} + s_j$ and $D_j = 2^{s_j} - 1$. Choose the initial index large enough that $s_j \geq 4$ and all these choices exist. Then

$$M_j 2^{s_j} \leq (\log(n_{j-1} + 3))^{3\varepsilon/4} \leq (\log(n + 3))^\varepsilon$$

at each coordinate n of the triple. This bound is also at most n after increasing the initial index. The moduli are again nested and tend through factorials of unbounded order, and the linear bound still gives summability. The congruence induction and telescoping estimate therefore apply to this schedule too. Now $s_j \asymp_\varepsilon \log \log n_j$, so an interval $[X, 2X)$ contains $O_\varepsilon(X / \log \log X)$ permitted indices.

We couple the two block distributions by choosing the same starting index. A changed coordinate belongs to at most m of the sampled blocks, so the probability that the blocks differ is at most $m|S \cap [X, 2X + m)|/X$. This proves the stated total variation

bound. When $m \leq X$, the interval $[X, 2X + m)$ is contained in $[X, 4X)$, and the support estimate on two dyadic intervals bounds the probability by $O_\varepsilon(m / \log \log X)$. The required uniform convergence follows. $\square$

The coupling also applies to tests depending on the starting index. If their absolute value is bounded by B_0 , their mean changes by at most $2B_0m|S \cap [X, 2X + m)|/X$. For a function of the block alone, the bound is $2B_0d_{TV}$. These are absolute errors: they give no relative estimate for an event whose probability tends to zero, nor a comparison of the complete infinite tails. The companion's [Section 2](#) also distinguishes upper Banach density from ordinary density.

3 Application to prime gaps

The elementary bound $p_n \leq 1250(n+1)^4$ ensures convergence of the prime and gap series. For completeness, the central-binomial argument of Erdős [19, pp. 194–196] gives, for $m \geq 4$,

$$4^m < m \binom{2m}{m} \leq m(2m)^{\pi(2m)}.$$

The left inequality follows by induction. For the right inequality, the full power of any prime in $\binom{2m}{m}$ is at most $2m$. If $m = (n+5)^4$ and $\pi(2m) \leq n$, these inequalities would give $4^m < m(2m)^n \leq 4^m$: with $x = n+5$, use $x \leq 2^x$ and $n+4(n+1)x \leq 2x^4$ for the last inequality. Thus $p_n \leq 2(n+5)^4 \leq 1250(n+1)^4$. The full prime-power calculation is in [Appendix A of the companion](#).

Corollary 3.1 (a rational sum with the stated prime-gap statistics). *Let $p_0 = 2, p_1 = 3, \dots$ be the primes and $g_n = p_{n+1} - p_n$. Given $K \in \mathbb{N}$ and $0 < \varepsilon \leq 1$, there is $b : \mathbb{N} \rightarrow \mathbb{N}$ with rational dyadic sum such that $b_n = g_n$ for $n < K$, $b_n \geq g_n$, and $b_n - g_n \leq (\log(n+3))^\varepsilon$ eventually. For every fixed positive modulus, both the coefficients and the cumulative positions eventually retain their corresponding residues. The empirical distributions of unnormalised blocks have total variation distance tending to zero for lengths $o(\log \log X)$, and for every fixed nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$,*

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$$|\{n < N : F(b_n, \dots, b_{n+k}) = 0\}| = o(N).$$

Moreover, with $P_n = 2 + \sum_{i < n} b_i$,

$$0 \leq P_n - p_n = O_\varepsilon\left(\frac{n(\log(n+3))^\varepsilon}{\log \log n}\right), \quad P_n \sim n \log n.$$

These positions are not asserted to be prime.

Lean proves this for every $\varepsilon > 0$, assuming Schlage-Puchta's Lemma 4 [18] only for the nonconcentration bound and the prime number theorem only for $P_n \sim n \log n$.

Proof. Since the gap series converges, we can apply Proposition 1.1 to $a = g$. Choose a rational r in its nondegenerate interval I and put $b = g + e$. The proposition gives all the correction and congruence assertions, together with the comparison of growing blocks. Schlage-Puchta [18, Lemma 4] proved that the zero set of each fixed nonzero polynomial in a fixed block of prime gaps has density zero. Outside $\bigcup_{i=0}^k (S - i)$ the two

blocks are equal, so a zero of $F(b_n, \dots, b_{n+k})$ is already a zero for the original gaps. The additional exceptional set has density zero, being a finite union of translates of S . This proves the required count without estimating the values of F .

To estimate $P_n - p_n = \sum_{i < n} e_i$, first observe that $|S \cap [0, n)| = O_\varepsilon(n / \log \log n)$. Indeed, the indices below $\sqrt{n}$ contribute at most $\sqrt{n}$ points, and on $[\sqrt{n}, n)$ we sum the dyadic support bounds, whose lengths add to $O(n)$ and whose $\log \log X$ are comparable to $\log \log n$. Multiplying by the pointwise correction bound gives the stated error. For $0 < \varepsilon \leq 1$ this is $o(n \log n)$, so the prime number theorem [12, Chapter 6] gives $P_n \sim n \log n$. Finally, summing nonnegative terms in the opposite order, we obtain

$$\sum_{n \geq 0} \frac{P_n}{2^{n+1}} = 2 + \sum_{i \geq 0} b_i \sum_{n > i} 2^{-n-1} = 2 + \sum_{i \geq 0} \frac{b_i}{2^{i+1}} = 2 + r \in \mathbb{Q}.$$

□

For $\varepsilon > 1$ we may use exponent 1, which gives smaller corrections and the error $O(n \log(n+3) / \log \log n)$. Thus the conclusions extend to every $\varepsilon > 0$. For each fixed $y \geq 2$, eventual congruence modulo $y!$ also gives $\gcd(P_n, y!) = 1$ once $p_n > y$. The cutoff depends on y , so this excludes no growing range of possible prime divisors. Even a prime-valued sequence of positions could omit primes.

Land's draft [8, Theorem 2] proves conditional irrationality under a uniform Hardy–Littlewood hypothesis of the kind formulated by Kuperberg [7, Conjecture 1.3]. Ringer's draft [16] obtains conditional normality. Their quantitative prime-pattern hypotheses are stronger than the absolute block comparison above and are not assumed here. The companion's Section 3 compares the hypotheses and sampling conventions.

4 The prime series and its actual tails

We return to the actual primes and write $G = \sum_{n \geq 0} g_n 2^{-(n+1)}$. To use a recurrence for these sums, we must retain the endpoint in finite summation by parts and then show that it vanishes. The polynomial bound from Section 3 supplies this limit and convergence of both series.

Theorem 4.1 (prime-to-gap identity). *The actual prime and gap series satisfy $\Pi = 2 + G$. Their complete tails*

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$$T_N = \sum_{j \geq 1} g_{N+j} 2^{-j}$$

satisfy $T_0 = 2G - 1$ and $T_{N+1} = 2T_N - g_{N+1}$. Thus Π , G and T_0 have the same rationality status.

Proof. Finite summation by parts gives

$$\sum_{i=0}^n \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}}.$$

Since $p_n = O((n+1)^4)$, the endpoint tends to zero and we obtain $\Pi = 2 + G$. We obtain the tail recurrence by splitting off the first term of T_N ; removing $g_0 = 1$ from G gives $T_0 = 2G - 1$. □

Tao noted this reduction in the problem's forum discussion [15, 7 October 2025].

Corollary 4.2 (exact irrationality reformulation). *The prime-value dyadic series is irrational if and only if the prime-gap dyadic series is. Both series converge by the polynomial bound proved above; neither side is proved irrational.*

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Proof. Adding the rational number 2 preserves rationality and irrationality. $\square$

The recurrence alone does not identify the infinite tails: adding $C2^N$ to any solution gives another solution. We distinguish the actual tails by the vanishing condition in the next lemma.

Lemma 4.3 (the boundary condition identifying a true tail). *Let $\sum_{j \geq 1} |a_j| 2^{-j} < \infty$ and $U_{N+1} = 2U_N - a_{N+1}$. Then $U_N = \sum_{j \geq 1} a_{N+j} 2^{-j}$ for every N if and only if $2^{-N} U_N \rightarrow 0$.*

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Proof. Iteration gives $2^{-N} U_N = U_0 - \sum_{j=1}^N a_j 2^{-j}$. The limit is zero exactly when U_0 is the full series. Subtracting its first N terms and multiplying by 2^N then gives the tail formula. Conversely the formula implies $2^{-N} U_N = \sum_{k > N} a_k 2^{-k} \rightarrow 0$ by absolute convergence. $\square$

4.1 Integral shifts

For the recurrence $U_{N+1} = 2U_N - a_{N+1}$ with $a_n \in \mathbb{Z}$, we write $D_h(N) = U_{N+h} - U_N$. Subtracting the integer a_{N+1} has no effect modulo $\mathbb{Z}$, so after N steps we have the same fractional part as $2^N U_0$. Integral shifts are consequently determined by the denominator of U_0 , independently of the coefficient sequence.

Theorem 4.4 (exact rationality classification). *For a real integer-coefficient recurrence, the following are equivalent: $U_0 \in \mathbb{Q}$; $D_h(N) \in \mathbb{Z}$ for some $h \geq 1, N \geq 0$; and, for some fixed $h \geq 1$, $D_h(N) \in \mathbb{Z}$ at every sufficiently large N . More precisely, if $U_0 = u/(2^s d)$ is in lowest terms, with d odd, then*

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$$\text{den}(U_N) = 2^{\max(s-N, 0)} d, \quad D_h(N) \in \mathbb{Z} \iff N \geq s \text{ and } d \mid 2^h - 1 \quad (h \geq 1).$$

Consequently U_0 is irrational exactly when every positive shift is nonintegral at every index, equivalently when for each fixed $h \geq 1$ there are arbitrarily late nonintegral shifts.

Proof. Iteration gives

$$U_N - 2^N U_0 \in \mathbb{Z}, \quad D_h(N) - 2^N (2^h - 1) U_0 \in \mathbb{Z}.$$

An integral shift with $h \geq 1$ makes the nonzero integer multiple $2^N (2^h - 1) U_0$ integral, and therefore forces $U_0 \in \mathbb{Q}$. Conversely, write $U_0 = u/(2^s d)$ in lowest terms. Multiplication by 2^N cancels exactly $\min(N, s)$ powers of two from the denominator. The remaining factor $2^h - 1$ is odd, so the second relation is integral exactly when $N \geq s$ and $d \mid 2^h - 1$. For $d > 1$, Euler's congruence provides such a shift $h = \varphi(d)$, and for $d = 1$ we take $h = 1$. This shift works for every $N \geq s$. Negating the pointwise and eventual criteria gives the two assertions about irrationality. $\square$

The theorem requires neither a true-tail boundary condition nor prime-gap coefficients. The eventual periodicity is that of fractional parts, and need not hold for the full tails. For instance, the positive even coefficients $2, 4, 2, 4, \dots$ starting at $n = 1$ give tails $8/3, 10/3$ alternately. Every shift of length 1 is nonintegral and every shift of length 2 is zero. This explains why a single shift length is insufficient.

5 Small tail differences and finite tests

Fix $h \geq 1$ and put $D_N = T_{N+h} - T_N$ and $\delta_N = g_{N+h+1} - g_{N+1}$. Then $D_{N+1} = 2D_N - \delta_N$, where δ_N is even. Two differences of absolute value less than one can both be integral only if they vanish, in which case $\delta_N = 0$. More precisely, we have the following result.

Proposition 5.1 (two consecutive differences of absolute value less than one). *Let $D, D' \in \mathbb{R}$, $\delta \in 2\mathbb{Z}$ and $D' = 2D - \delta$. The conditions $|D| < 1$, $|D'| < 1$ and $\delta \neq 0$ hold exactly when, for some $s \in \{-1, 1\}$,*

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$$\delta = 2s, \quad \frac{1}{2} < sD < 1.$$

In that case $sD' \in (-1, 0)$ and both D and D' are nonintegral.

Proof. The bounds on D, D' imply $-3 < \delta < 3$, so a nonzero even δ equals $2s$ with $s \in \{-1, 1\}$. Substituting in $D' = 2D - 2s$ gives $1/2 < sD < 1$. Conversely this interval gives $sD' = 2sD - 2 \in (-1, 0)$. $\square$

For $\delta = 2$, the endpoints $D = 1/2, 1$ give $D' = -1, 0$ respectively. The zero triple $D = D' = \delta = 0$ shows why the mismatch is required.

Problem 5.2 (two small tail differences at arbitrarily large indices). For every integer $h \geq 1$ and every cutoff N_0 , exhibit $N \geq N_0$ with

$$|T_{N+h} - T_N| < 1, \quad |T_{N+h+1} - T_{N+1}| < 1, \quad g_{N+h+1} \neq g_{N+1}. \quad (3)$$

A solution would give a nonintegral shift arbitrarily late for every h , and hence irrationality by Theorem 4.4. All three conditions must hold at the same index; witnesses for the separate conditions cannot be combined without further information. Schlage-Puchta's lemma [18, Lemma 4], applied to $x_h - x_0 \pm 2$, shows that the eligible indices with $\delta_N = \pm 2$ already have density zero. Separate infinite sets of small-difference and mismatch witnesses would therefore not suffice.

5.1 Finite separation from the integers

To decide nonintegrality from finitely many gaps, we bound the part of the difference that has not been summed. Choose $M(n) \geq g_n$ with $\sum_n M(n)2^{-n} < \infty$, and write

$$S_{h,N,L} = \sum_{j=1}^L (g_{N+h+j} - g_{N+j})2^{-j}, \quad R_{h,N,L}(M) = \sum_{j>L} (M(N+h+j) + M(N+j))2^{-j}.$$

Then $|D_N - S_{h,N,L}| \leq R_{h,N,L}(M)$. A general real majorant need not give an effective remainder bound, but the polynomial choice below does.

Proposition 5.3 (finite separation criterion). *If for every $h \geq 1$ and every cutoff N_0 there are $N \geq N_0, L \geq 1$ with*

$$\text{dist}(S_{h,N,L}, \mathbb{Z}) > R_{h,N,L}(M), \quad (4)$$

then Π is irrational.

Proof. The enclosure gives $\text{dist}(D_N, \mathbb{Z}) \geq \text{dist}(S_{h,N,L}, \mathbb{Z}) - R_{h,N,L}(M) > 0$. Thus each fixed positive shift is nonintegral at arbitrarily large indices, and Theorem 4.4 applies. $\square$

Remark (what arbitrary truncation does not gain). Whenever $|D - S_L| \leq R_L \rightarrow 0$, nonintegrality of D is equivalent to $\text{dist}(S_L, \mathbb{Z}) > R_L$ for some L . Indeed, for $\delta = \text{dist}(D, \mathbb{Z}) > 0$, take $R_L < \delta/2$ and use the 1-Lipschitz property of distance. Unrestricted depth therefore gives an exact reformulation, not an additional analytic estimate. At a prescribed logarithmic depth one must also control the margin from the integers as N varies; a shrinking remainder alone does not do so. The companion, Appendix D.2, gives an explicit example distinguishing refinement at a fixed value from enclosures of changing values.

We obtain a computable bound by taking $M(n) = 1250(n+2)^4$. For $P(x) = x^4 + 8x^3 + 36x^2 + 104x + 150$, summing the identity $2P(x) = (x+1)^4 + P(x+1)$ with dyadic weights gives $\sum_{j \geq 1} (x+j)^4 2^{-j} = P(x)$ and hence

$$R_{h,N,L}(M) = 1250 2^{-L} (P(N+h+L+2) + P(N+L+2)).$$

If $\delta_N = 2s$ and $|QD_N - A| \leq B$, with $A \in \mathbb{Z}, Q > 0, B \geq 0$ integers, the inequalities

$$2sA - Q > 2B, \quad Q - sA > B$$

put sD_N in $(1/2, 1)$. The numerator A is signed. At $h = 1, N = 2, L = 40$, one has $\delta_2 = g_4 - g_3 = -2$ and

$$Q = 2^{40}, \quad A = -662838684750, \quad B = 11764181250, \quad s = -1.$$

Here $A/Q = S_{1,2,40}$ and $B/Q = R_{1,2,40}(M)$, and the two strict margins are 202637379224 and 424908761776. The [companion's integer certificates](#) give the prime-index calculation and remainder derivation. This finite pair leaves the quantifiers in Problem 5.2 unresolved.

6 Further consequences and questions

We can place the interval in Proposition 1.1 inside $(A, A + \eta)$ for any $\eta > 0$. Choose $K' \geq K$ such that $(K' + 1)2^{-K'} < \eta$, and start the construction after K' . The bound $e_n \leq n$ from the proof gives Then

$$0 < \sum_n e_n 2^{-(n+1)} \leq (K' + 1)2^{-K'} < \eta.$$

This bounds the dyadic change, without bounding the ordinary total.

6.1 Nonperiodic coefficients

The usual periodicity criterion for rational binary expansions assumes digits in $\{0, 1\}$. With unrestricted integer coefficients, the following telescoping identity already gives a different behaviour:

$$\sum_{i=0}^{n-1} \frac{i-1}{2^{i+1}} = -\frac{n}{2^n} \rightarrow 0.$$

Its coefficients are unbounded and its first coefficient is -1 . We obtain a positive even example by taking $c_n = 2(n^2 + 4n + 2)$ and $U_n = 2(n + 4)^2$. Direct substitution gives $U_{n+1} = 2U_n - c_{n+1}$, while the polynomial growth gives $2^{-n}U_n \rightarrow 0$. Lemma 4.3 therefore identifies U as the complete tail and yields

$$\sum_{j \geq 1} c_j 2^{-j} = 32, \quad \sum_{n \geq 0} c_n 2^{-(n+1)} = 18.$$

The coefficients increase strictly and $c_{n+1} - c_n = 4n + 10$ is never ± 2 . The companion's Section 8 gives further examples, including a bounded nonperiodic sequence in Appendix D. None of these examples constructs consecutive primes.

Proposition 6.1 (prime gaps do not become periodic). *For every positive h , the actual consecutive-prime-gap sequence is not eventually periodic with period h .*

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Proof. If the gaps were eventually periodic, they would be bounded. For each $m \geq 2$, however, the $m - 1$ integers $m! + 2, \dots, m! + m$ are composite. The consecutive primes on either side therefore have a gap of at least m , contradicting boundedness. $\square$

The results on bounded prime gaps [13], clusters of each fixed size [11, 20], and extreme large gaps [4] leave the signed weighted-tail windows above uncontrolled. This remains so with Polymath's bound $\liminf_n g_n \leq 246$ [20, Theorem 1.4(i)]. At a logarithmic truncation depth the explicit remainder tends to zero, but the finite sum must still be separated from the relevant boundary by a larger margin. For the actual prime series, nonintegral shifts are needed arbitrarily late for every positive length. Problem 5.2 would supply them. The perturbation theorem concerns the insufficiency of the specified congruences and block statistics, and leaves possible arguments using primality or quantitative rare-event estimates untouched.

7 Sources and further comparisons

Formal proofs. A result with a kernel-checked Lean proof carries a mark in the margin. Lean opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. Comparator opens the record of an independent check, in which the same statement, written again from Mathlib alone in a separate repository, was compared with our proof by Lean's Comparator tool, allowing only the three standard axioms. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The evidence record gives every declaration, version and check. These checks show that the stated propositions

are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text. The linked Lean declarations use Lean 4 [9] and mathlib [10]. The formal proof of Proposition 1.1 changes one coordinate at each stage, whereas the proof here uses triples. Both fix the permitted set, the interval and the congruence cutoffs before choosing the target. For Corollary 3.1, the formal statement covers every $\varepsilon > 0$ and assumes Schläge-Puchta’s lemma only for nonconcentration and the prime number theorem only for the asymptotic $P_n \sim n \log n$. The source links record formal support for the linked statements, rather than verification of the different printed proofs. The companion’s Appendix G retains the complete source index at this note’s original pin, including the finite telescoping identities and their nonperiodicity consequences. Its Section 3 retains the broader subsum and automatic-sequence comparisons.

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